

Complex network analysis

- advanced analysis

Reference:
Chapter 11 and 12, Networks, by Mark Newman

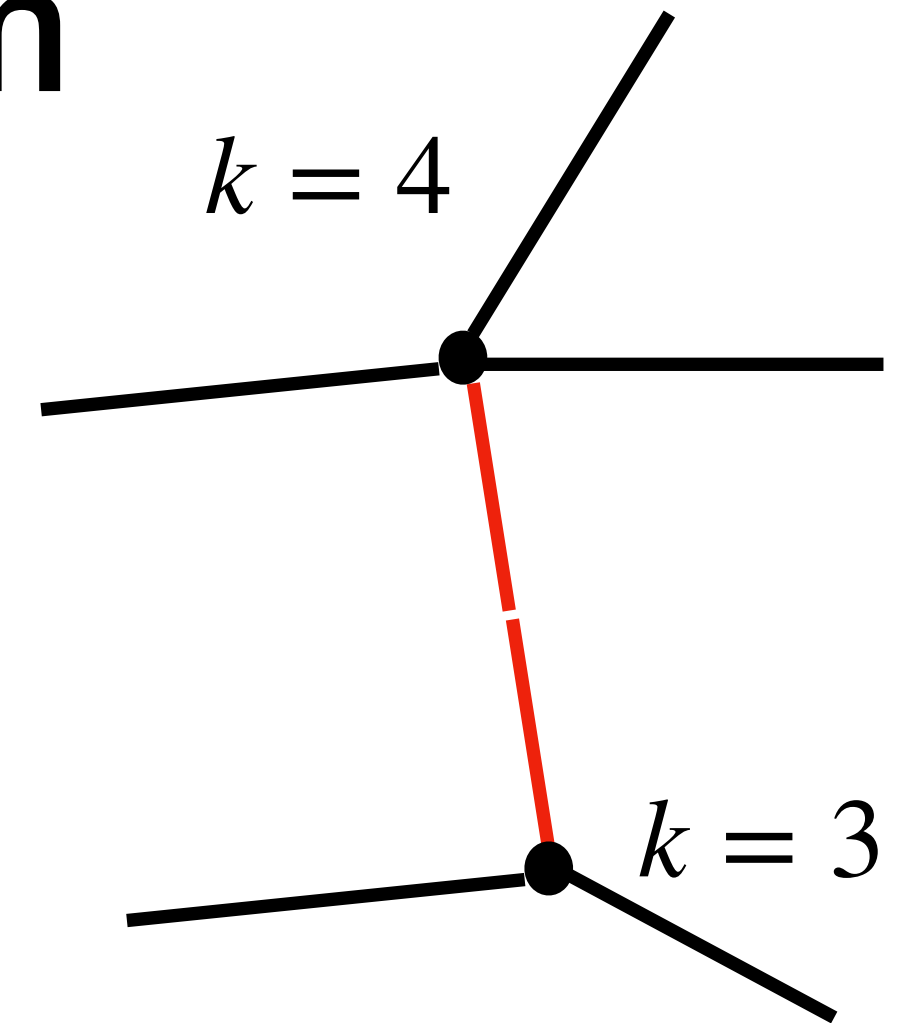
Friendship paradox & excess degree distribution

“Your friends have more friends than you do.”

Mathematical definition of the problem

Assumptions:

1. Everyone lives in the same social network: $\mathbf{G}$ with average degree $\langle k \rangle \ll n$ and m links
2. Assume the social network's can be mostly described by the degree distribution p_k
 - This also assumes links randomly connect nodes.
 - This is called the '*configuration model*' in Newman's book Chapter 12.



Quantity of interest: Average number of friends of my friends

What is the probability that a certain friend of mine has degree k ?

If a node has degree k , the chance that it is this friend of mine: $\frac{k}{2m}$

number of nodes with degree k : np_k

$$\frac{k}{2m} \times np_k = \frac{kp_k}{\langle k \rangle}$$

$$c = \langle k \rangle = 2m/n$$

Expectation of the above variable is the average degree of my friends:

$$\sum_k k \frac{kp_k}{\langle k \rangle} = \frac{\langle k^2 \rangle}{\langle k \rangle}$$

What is the difference between this average and $\langle k \rangle$?

$$\frac{\langle k^2 \rangle}{\langle k \rangle} - \langle k \rangle = \frac{1}{\langle k \rangle} (\langle k^2 \rangle - \langle k \rangle^2) = \frac{\sigma_k^2}{\langle k \rangle}$$

Real world numbers

$$\sigma_k^2 / \langle k \rangle > 0$$

Average degree of neighbours is always larger than average degree of the network

For scale-free network, this difference diverges since variance diverges !!!

That's why your friends' appear to have more friends than you!

Any exceptions to this?

Remember we assumed random connections, which are not realistic.

But such estimation holds up well in real networks:

Network	n	Average degree	Average neighbor degree	$\frac{\langle k^2 \rangle}{\langle k \rangle}$
Biologists	1 520252	15.5	68.4	130.2
Mathematicians	253339	3.9	9.5	13.2
Internet	22963	4.2	224.3	261.5

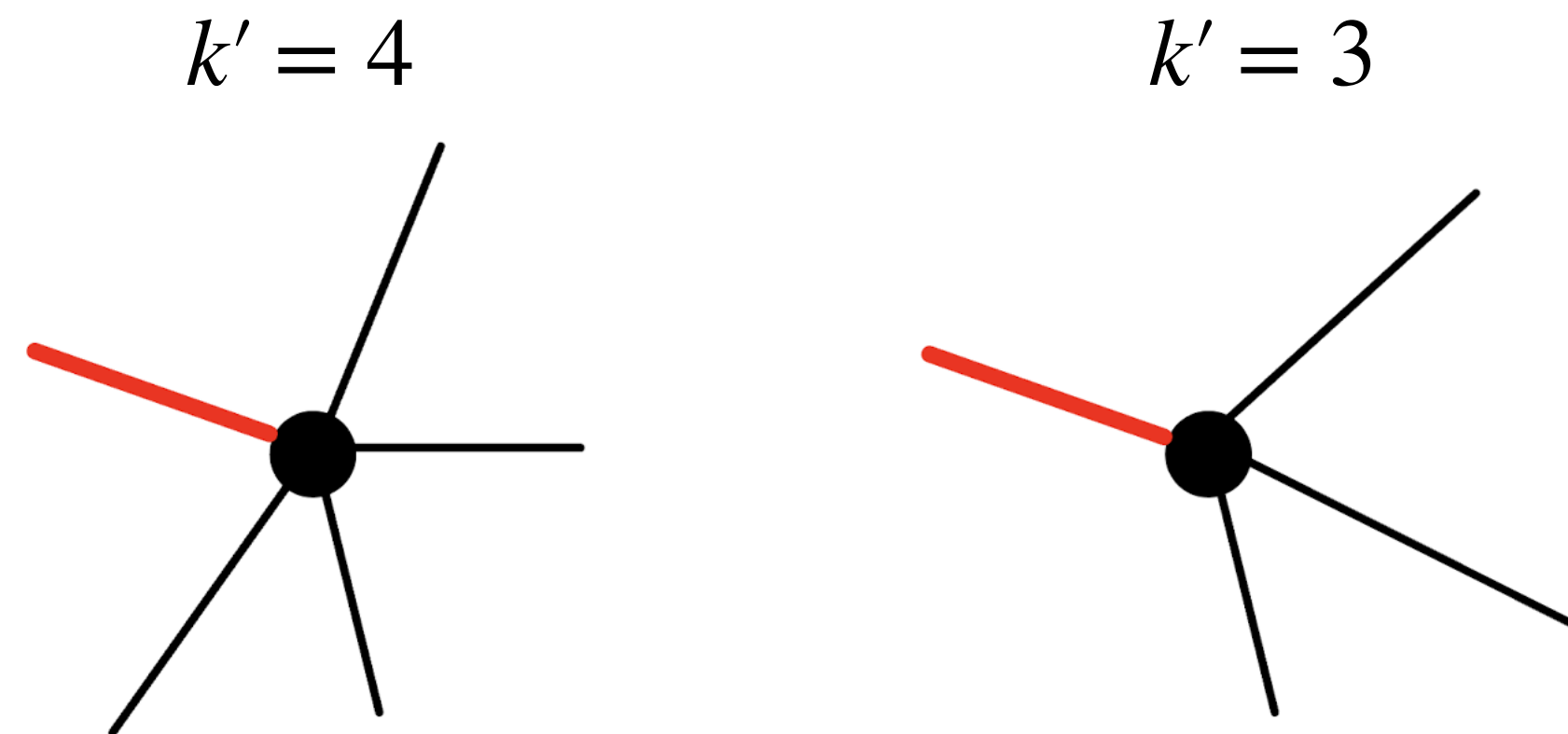
Excess degree distribution

*“In most of the calculations that follow, we will be interested not in the total degree of the node at the end of an edge but in the number of edges attached to that node other than the one we arrived along (or how many links my friend has excluding myself). This number is called the **excess degree** of the node and it is just equal to the total degree minus one.*

p_k is the degree distribution,
then the excess degree distribution $q_{k'}$ is

$$q_{k'} = \frac{k' + 1}{2m} \times np_{k'+1} = \frac{(k' + 1)p_{k'+1}}{\langle k \rangle}$$

$$\langle k \rangle = c_1 = 2m/n$$



This way, we turn the perspective to ‘**edge**’ centric rather than node centric.

For many problems, we can repeatedly follow the edges to find out the mathematical relations.

(Note: going forward, we may ignore the ' in k' and use k instead for excess degree distribution notations.)

Clustering coefficient

The probability that two neighbours of a node are connected

Consider an arbitrary node v with two neighbours i and j

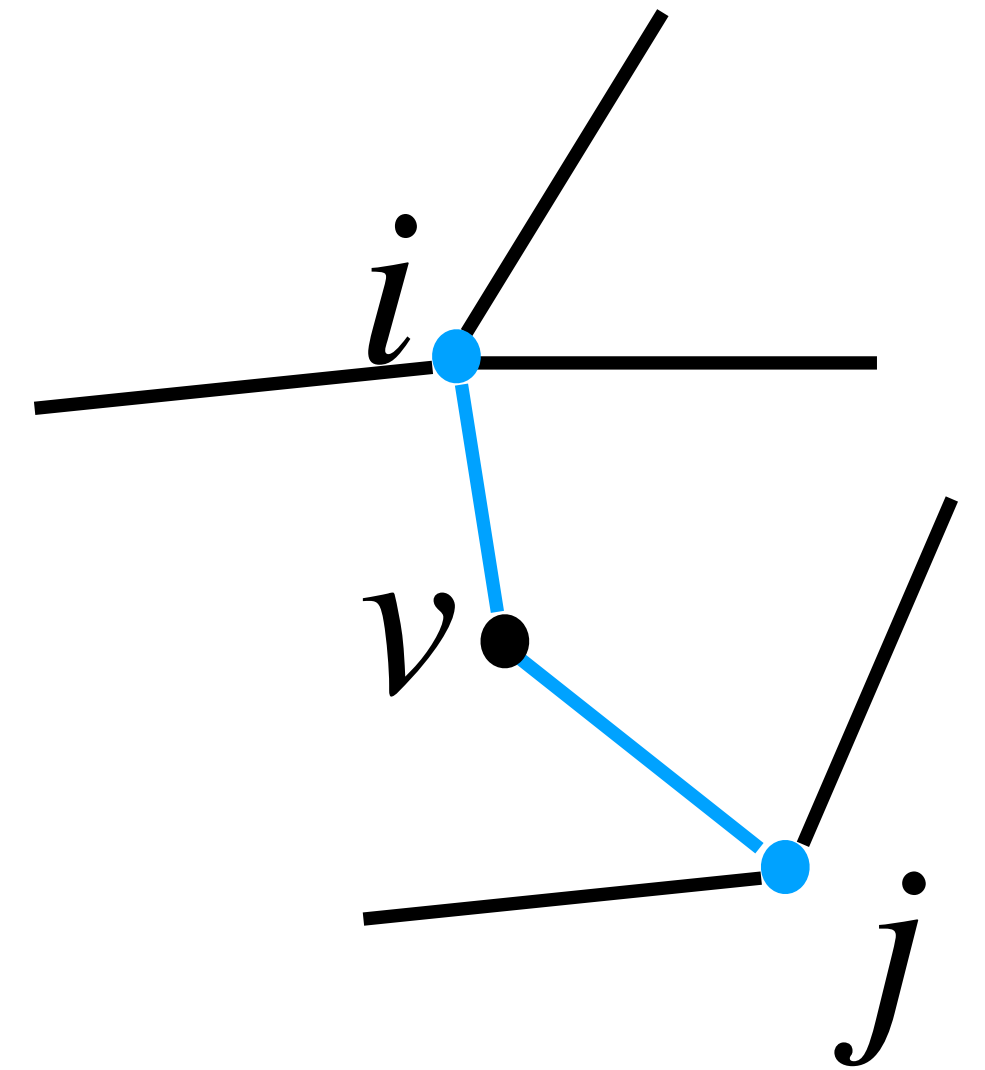
Prob that i has degree k_i

Prob that i and j are connected

Prob that j has degree k_j

i and j are independent

$$\begin{aligned}
 C &= \sum_{k_i, k_j=0}^{\infty} q_{k_i} q_{k_j} \frac{k_i k_j}{2m} = \frac{1}{2m} \left[\sum_{k=0}^{\infty} k q_k \right]^2 \\
 q_k &= \frac{(k+1)p_{k+1}}{\langle k \rangle} \\
 &= \frac{1}{2m \langle k \rangle^2} \left[\sum_{k=0}^{\infty} k(k+1)p_{k+1} \right]^2 = \frac{1}{2m \langle k \rangle^2} \left[\sum_{k'=1}^{\infty} (k'-1)k'p_{k'} \right]^2 \\
 &= \frac{1}{n} \frac{[\langle k^2 \rangle - \langle k \rangle]^2}{\langle k \rangle^3},
 \end{aligned}$$



We learnt for ER network with connection prob p , k is Poisson distribution

$$\sigma_k^2 = \langle k^2 \rangle - \langle k \rangle^2 = \langle k \rangle \longrightarrow \langle k^2 \rangle - \langle k \rangle = \langle k \rangle^2$$

$$C = \frac{1}{n} \frac{\langle k \rangle^{2 \times 2}}{\langle k \rangle^3} = \frac{\langle k \rangle}{n} = p \approx 0 \text{ for } n \rightarrow \infty$$

For scale free networks, 2nd moment $\langle k^2 \rangle$ diverges

$C \rightarrow \infty$ Very large clustering coefficient!

Locally tree like structure

In the configuration model of a sparse network, i.e. all links are random, for a finite neighbourhood of any node, the connected nodes within the neighbourhood has to be tree structure

Argument:

- For a finite distance d , all the nodes that reachable from node v within distance d forms a sub graph with size s (also finite).
- Then this graph has at least $s - 1$ links (a tree).
- For any extra link (say connecting node i and j) in excess of the $s - 1$ links within this sub graph, the conditional probability (conditioned on this sub graph) that it exists is 0.

$$\frac{k_i k_j}{2m} = \frac{k_i k_j}{\langle k \rangle n} \rightarrow 0 \text{ for } n \rightarrow \infty$$

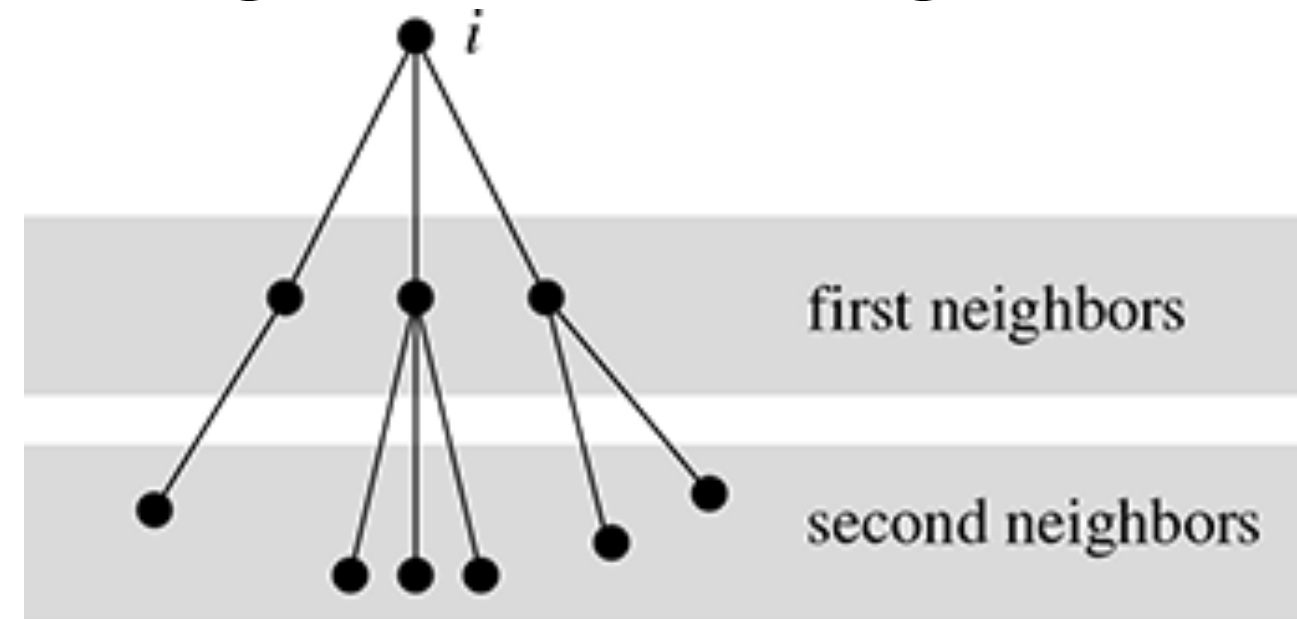
- Hence, the chance to have the extra link is almost 0 -> locally tree structure.

This tree like property is important for many of the following analysis.

Number of second neighbours

Total number of 2nd neighbours of node i :

Same as total excess degree of its neighbours - tree like property



Average excess degree is:

$$\langle k \rangle_{\text{excess}} = \sum_{k=0}^{\infty} k q_k = \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} k(k+1) p_{k+1} = \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} (k-1) k p_k = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}$$

Average number of 2_{nd} neighbors for node i is:

$$k_i \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}$$

Average number of 2_{nd} neighbors for all nodes in the network is:

$$c_2 = \langle k \rangle \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = \langle k^2 \rangle - \langle k \rangle$$

Again, depends on the 2_{nd} moment!

Number of d th neighbors

Average number of 1st neighbors (a.k.a. degree) is:

$$c_1 = \langle k \rangle$$

Average number of 2nd neighbors is:

$$c_2 = \langle k \rangle \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = \langle k \rangle \langle k \rangle_{\text{excess}}$$

Average number of 3rd neighbors is:

$$c_3 = c_2 \times \langle k \rangle_{\text{excess}} = \frac{c_2^2}{c_1}$$

Average number of d _{th} neighbors is:

$$c_d = \left(\frac{c_2}{c_1} \right)^{d-1} c_1$$

Average excess degree:

$$\langle k \rangle_{\text{excess}} = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = \frac{c_2}{c_1}$$

Giant component (GC)

In many real world networks, a ‘**sizeable**’ **fraction** of the nodes need to be connected to function

Internet

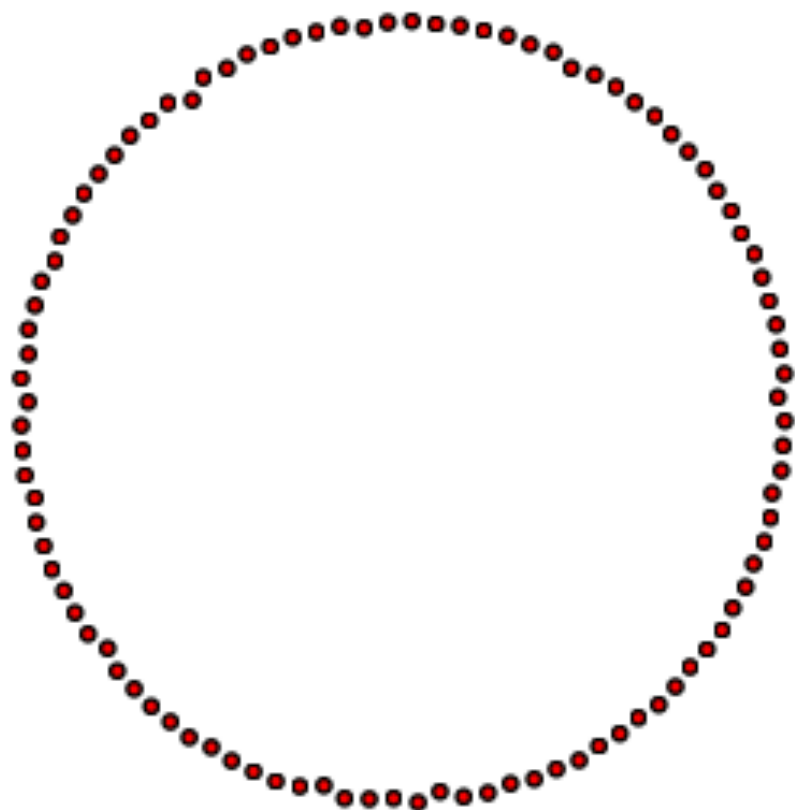
Road

Banking

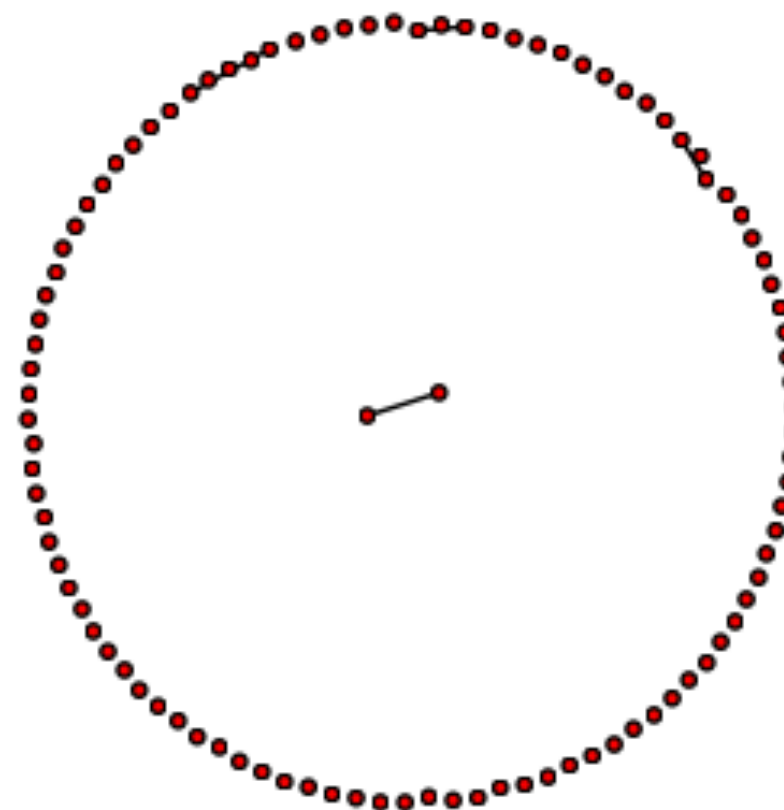
Power grids

Such ‘sizeable’ connected component is called: **GIANT COMPONENT**

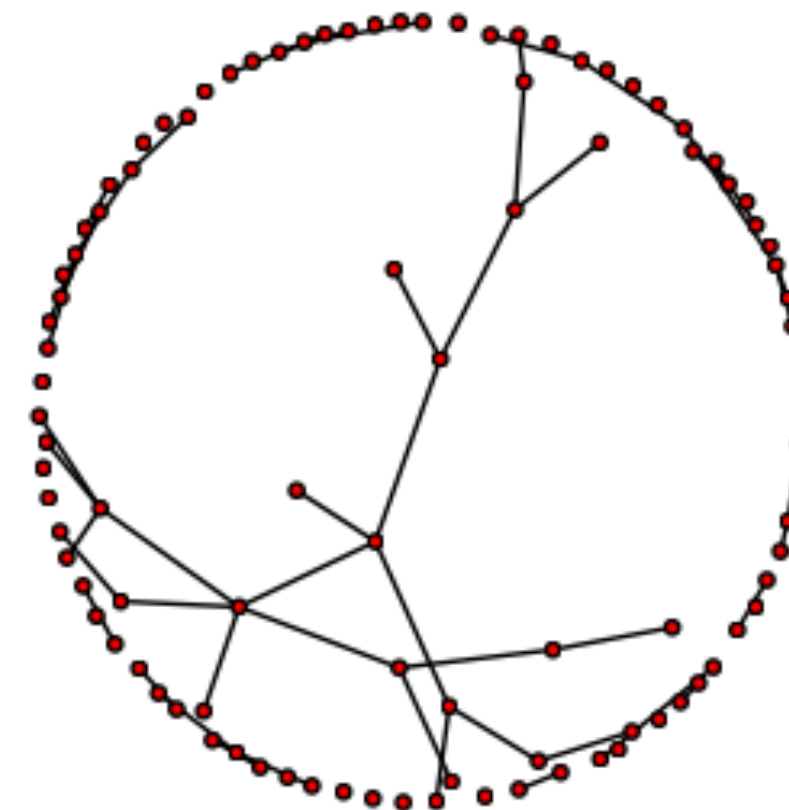
$p = 0.0001$



$p = 0.001$



$p = 0.01$



$p = 0.1$

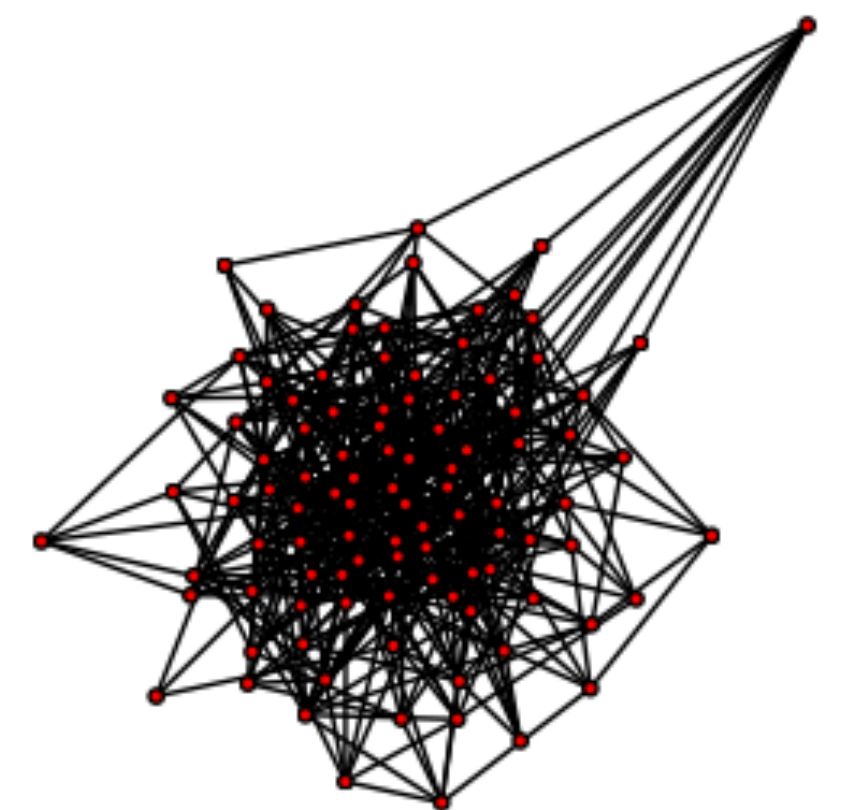


Illustration of giant component in Erdos Renyi network with different p values.

Giant component (GC)

Giant component is defined as :
the connected component that contains a ***finite*** fraction of all nodes in the network.

1. Giant component has to be unique:

There cannot be more than **ONE** giant component

Proof: assuming there are two giant components with size $s_1 = an$ and $s_2 = bn$ nodes.

The probability that there is NO link between them is:

$$Q_2 \approx \left(1 - \frac{c}{n}\right)^{an \cdot bn}$$
$$\downarrow \begin{aligned} \ln Q_2 &= abn^2 \ln \left(1 - \frac{c}{n}\right) \\ &\approx -abn^2 \frac{c}{n} \\ &= -abcn \end{aligned}$$

$$Q_2 \approx e^{-abcn} \approx 0 \text{ for } n \rightarrow \infty$$

Because $a, b, c > 0$ & *finite*

The chance of having 2 GC is 0! So GC has to be unique (if exists).

2. Phase transition phenomenon for giant component to emerge

There is a critical threshold, beyond which giant component will emerge

Probability that a link from a node does not lead to GC u

$k = 4$

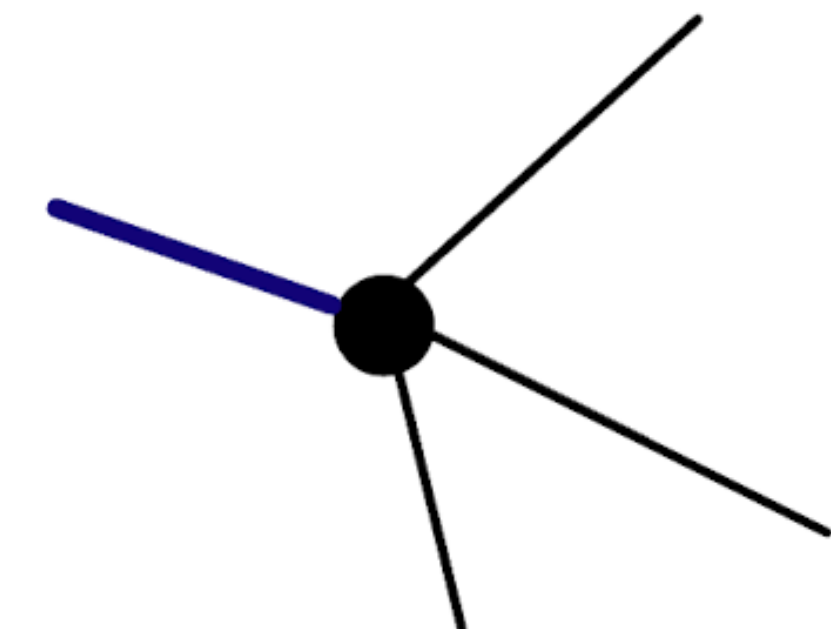
Probability that a node with k links does not lead to GC u^k

Average probability a node is not in the GC

$$g_0(u) = \sum_k p_k u^k$$

Average probability a node is in the GC
(This is also the fractional size of the GC)

$$\boxed{S = 1 - g_0(u)}$$



Pick a random link, follow it to reach a node

Probability that none of the other k' links
of the node is in the GC

$u^{k'}$

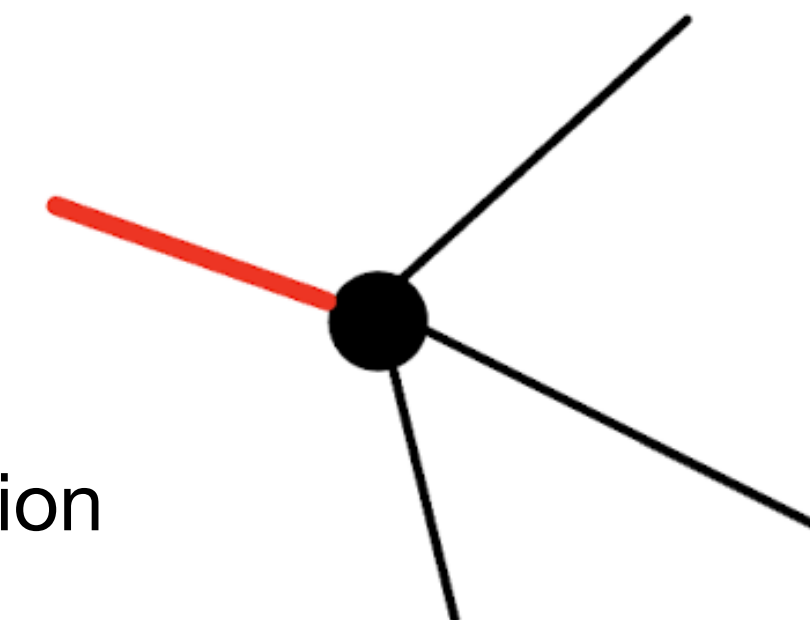
$k' = 3$

We can define similar function:

Probability that this link does not lead to GC is

$$g_1(u) = \sum_{k'} q_{k'} u^{k'}$$

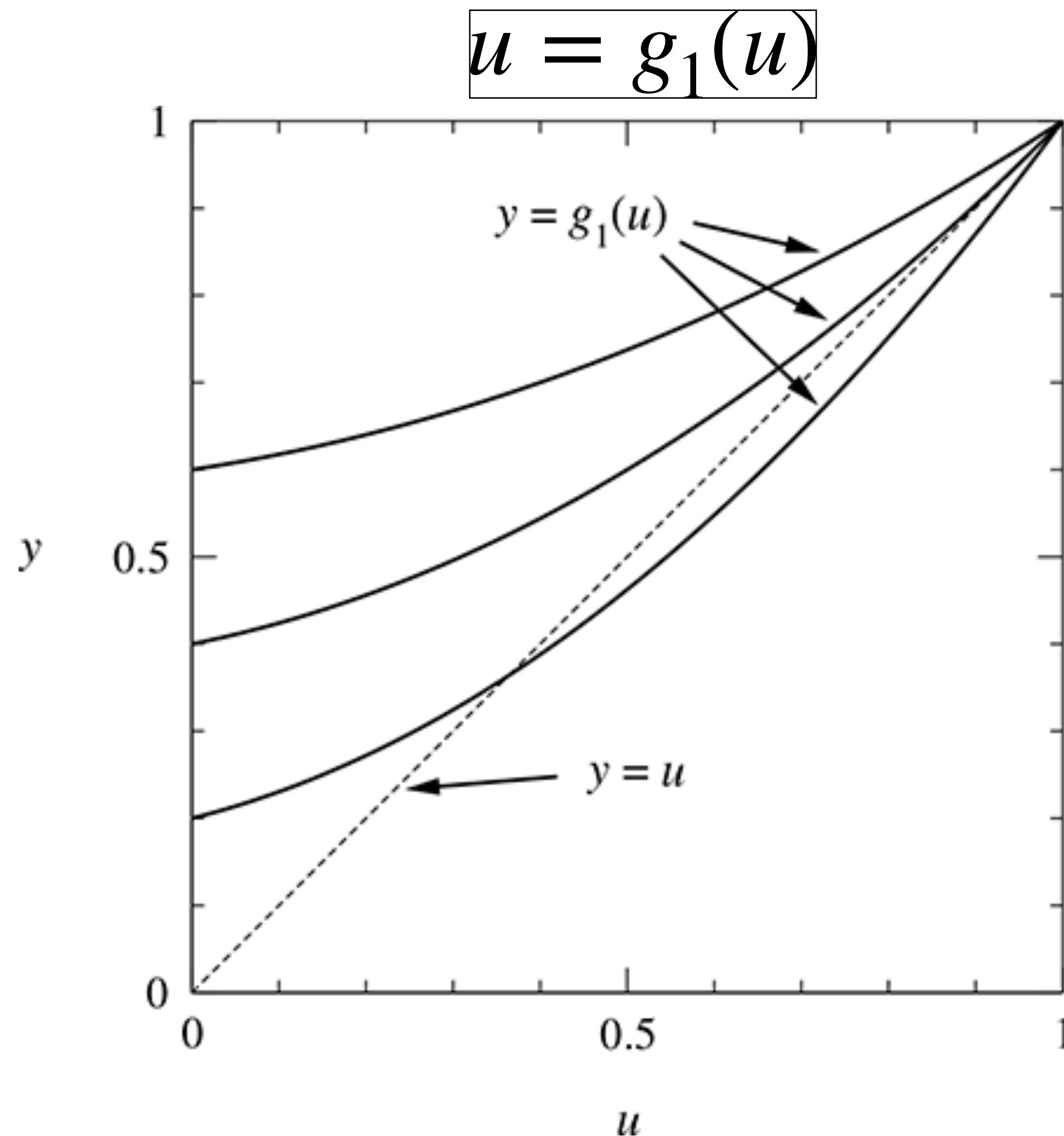
$\downarrow u = g_1(u)$ by definition



Condition for GC to appear: when a solution $u < 1$ exists for

$$\boxed{u = g_1(u)}$$

Solving for this transition



$$g_1(u) \big|_{u=1} = \sum_k q_k 1^k = 1$$

Graphically, the solution $u < 1$ appears when the curve $y = g_1(u)$ is tangential to the line $y = u$ at $u = 1$.

$$\frac{dg_1(u)}{du} \big|_{u=1} > 1$$

$$\frac{d}{du} \sum_k q_k u^k \big|_{u=1} > 1$$

$$\sum_k k q_k u^{k-1} \big|_{u=1} > 1$$

At $u = 1$:
Average excess degree

$$\frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} > 1$$

Simple explanation: if you follow a link, and every time when reaching a node, it will lead to at least one other node, you will eventually reach infinity number of nodes - Giant component.

Example on ER network

Condition for transition

$$\boxed{\frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} > 1} \longrightarrow \langle k^2 \rangle - 2\langle k \rangle > 0$$

Due to Poisson degree distribution: $\langle k^2 \rangle = \langle k \rangle^2 + \langle k \rangle$

Critical threshold is

(Recall $c_1 = \langle k \rangle$) $c_1^2 + c_1 - 2c_1 > 0$

$$c_1^2 - c_1 > 0$$

$$c_1 > 1$$

$$p > \frac{1}{n}$$

$$\boxed{p_c = \frac{1}{n}}$$

The critical threshold turns out to be trivially intuitive for a complex network:

$$\boxed{\langle k \rangle_{\text{excess}} = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} > 1} \longrightarrow \text{Average excess degree} > 1$$

On average, each node needs to extend out to at least another node to form a GC.

Example on ER network

Condition for transition

$$\boxed{\frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} > 1} \longrightarrow \langle k^2 \rangle - 2\langle k \rangle > 0$$

Due to Poisson degree distribution: $\langle k^2 \rangle = \langle k \rangle^2 + \langle k \rangle$

Critical threshold is

Recall $c_1 = \langle k \rangle$: $c_1^2 + c_1 - 2c_1 > 0$

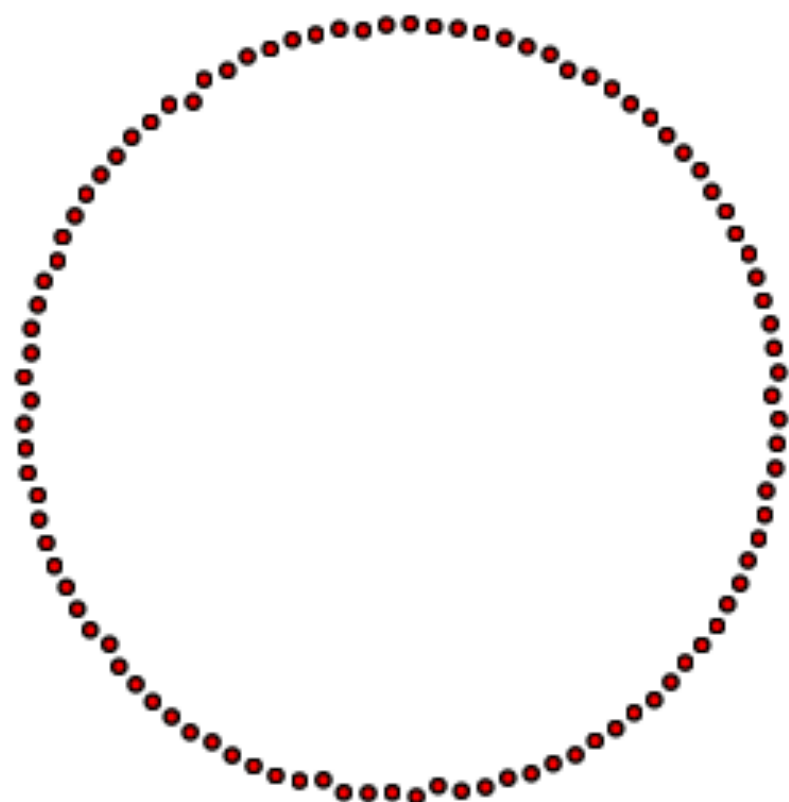
$$c_1^2 - c_1 > 0$$

$$c_1 > 1$$

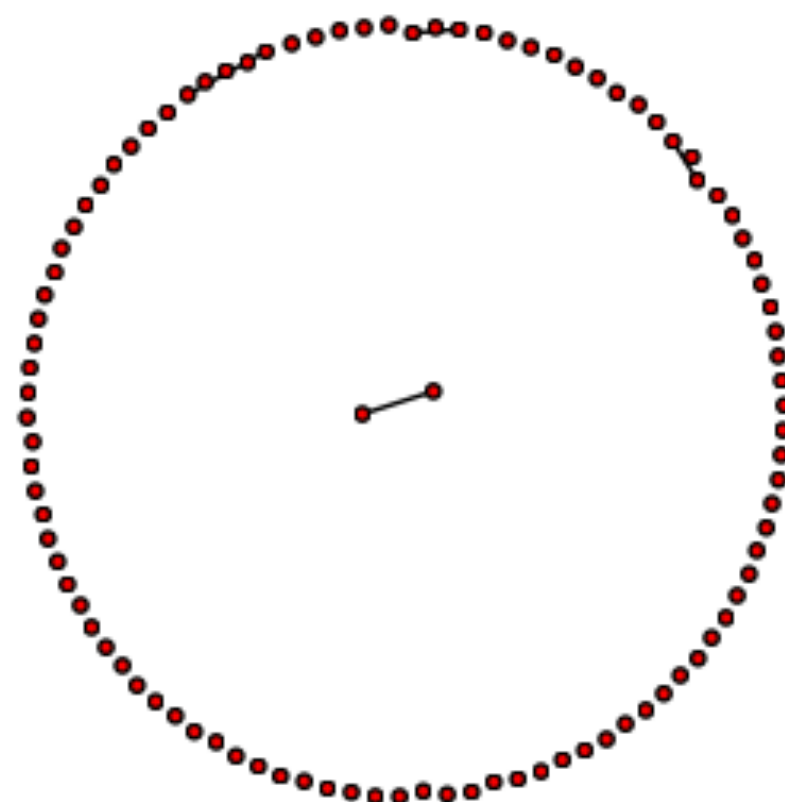
$$p > \frac{1}{n}$$

$$\boxed{p_c = \frac{1}{n}}$$

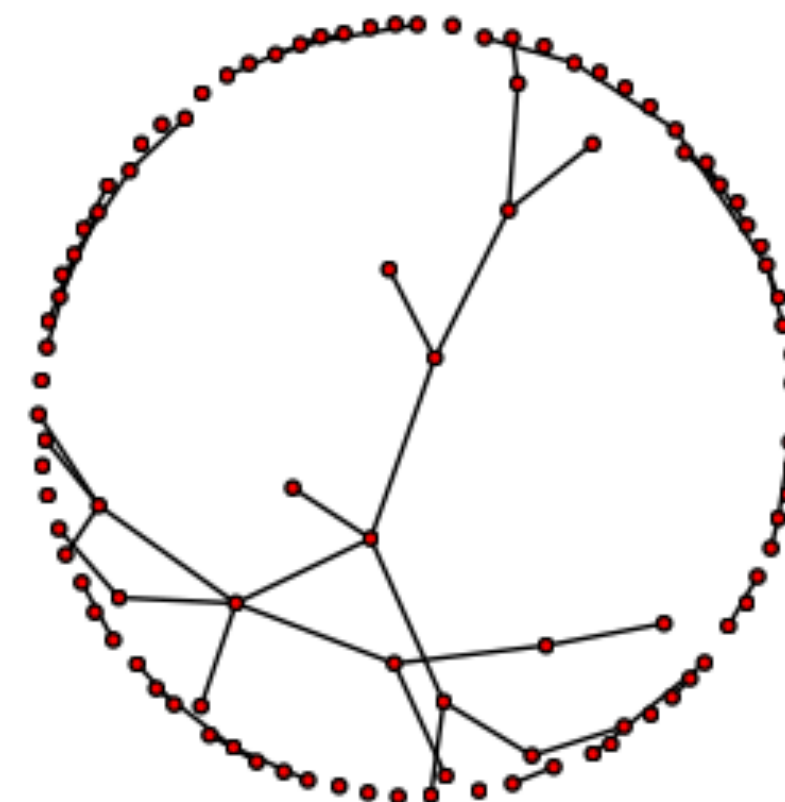
$p = 0.0001$



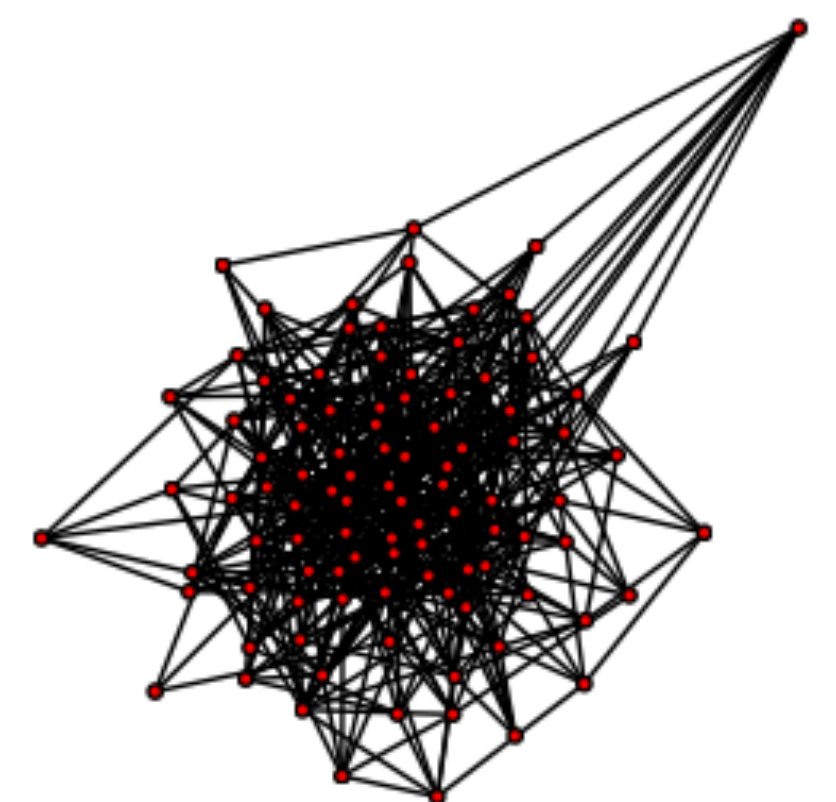
$p = 0.001$



$p = 0.01$

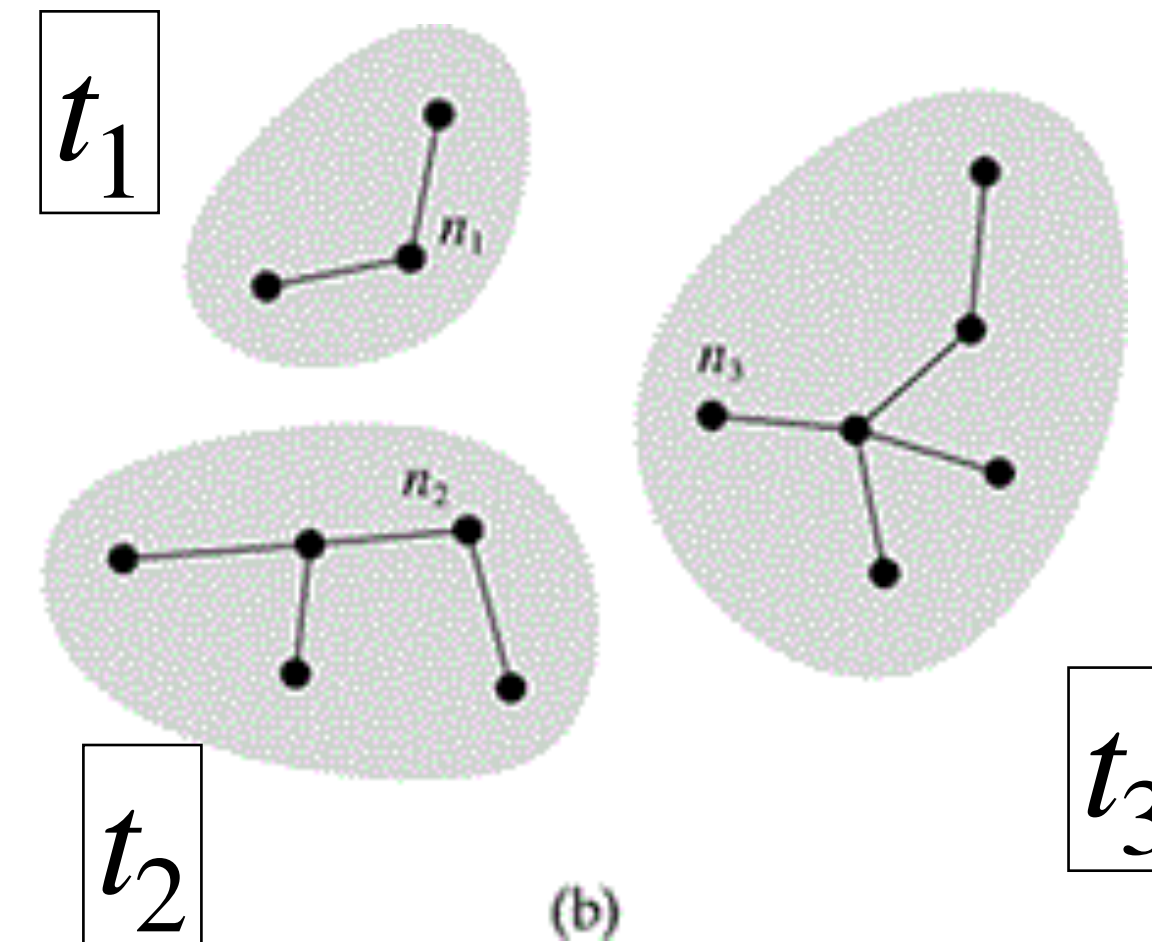
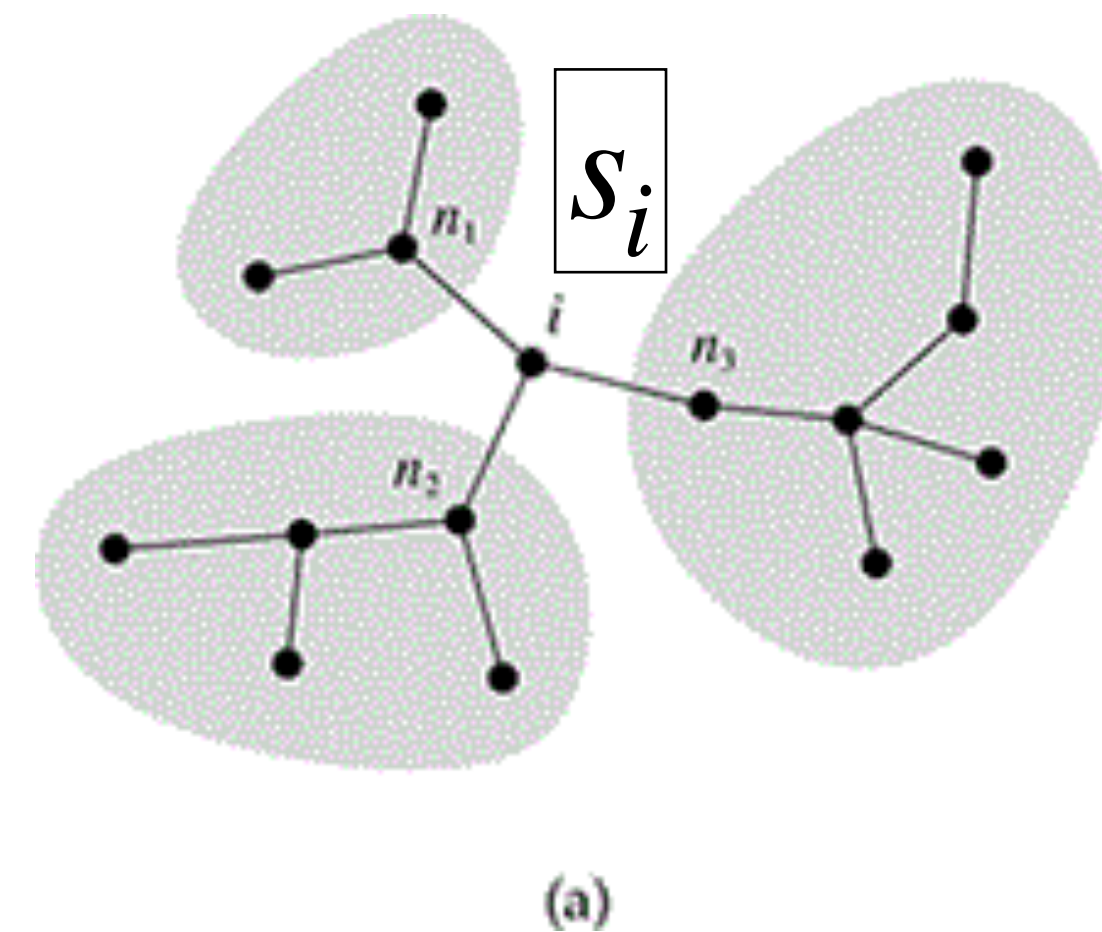


$p = 0.1$



Small component sizes

Small components are finite in sizes: diameter is finite $\rightarrow$ they are tree structure as proven before



s_i : the size of the small component **CONTAINING** node i

t_j : number of nodes **REACHABLE** from an edge (neighbor j) of node i
(Note: t_j is independent of i due to random assumption)

Average size of small components:

$$\langle s \rangle = 1 + \langle k \rangle_{\text{small}} \langle t \rangle$$

$\langle k \rangle_{\text{small}}$ is the average degree of nodes in small components

$\langle t \rangle$ is the average size of t

Given that a node is in a small component, its degree distributions is (Bayes rule):

$$P(\text{degree } k | \text{small component}) = P(\text{small component} | \text{degree } k) \frac{P(\text{degree } k)}{P(\text{small component})}$$

Unconditional degree distribution $P(\text{degree } k) = p_k$

Probability not in Giant component of size S

$$P(\text{small component}) = 1 - S = g_0(u)$$

Probability the node is in small component given that it has degree k :

$$P(\text{small component} | \text{degree } k) = u^k$$

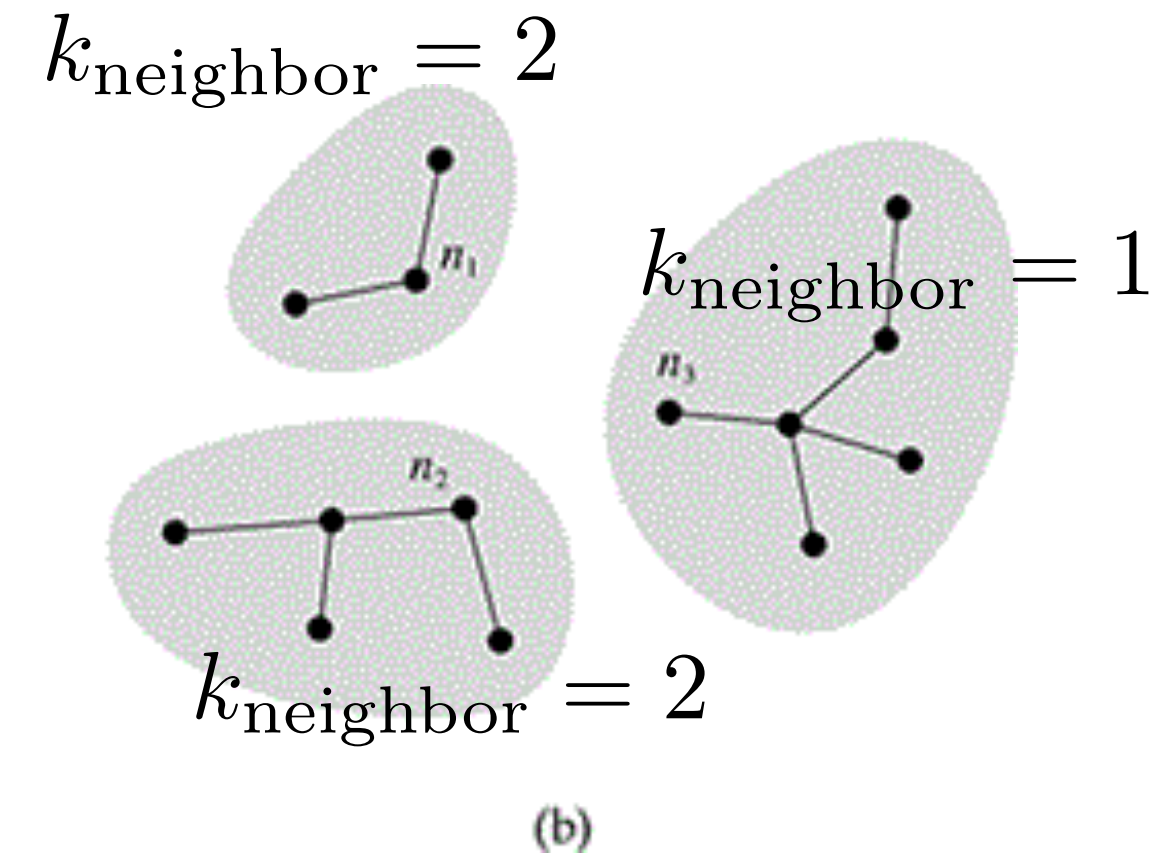
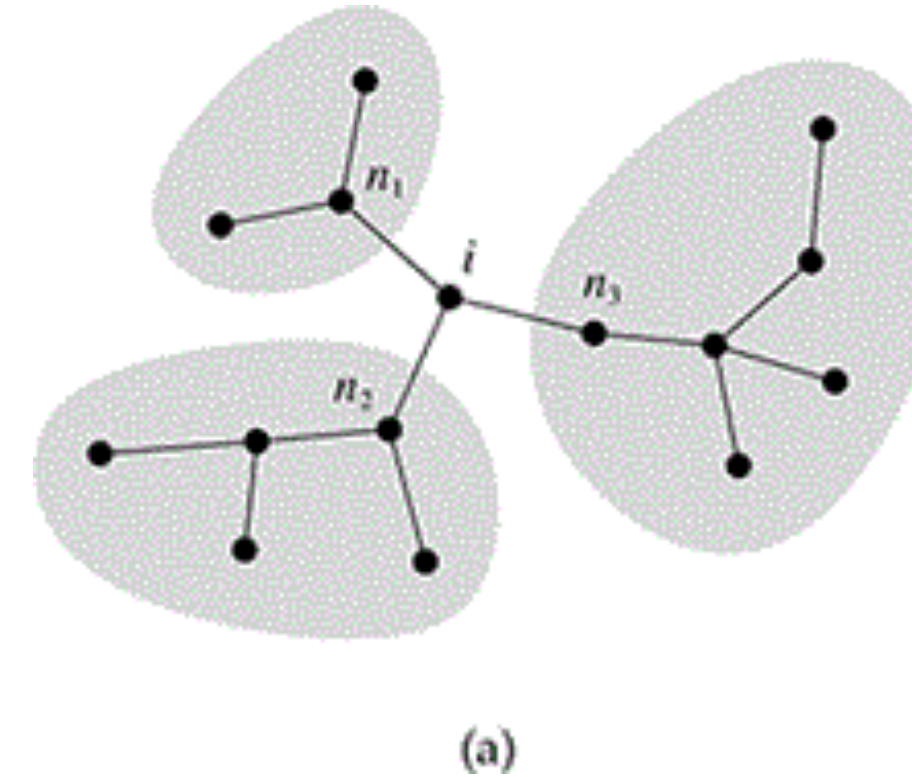
$$P(\text{degree } k | \text{small component}) = \frac{p_k u^k}{g_0(u)}$$

$\langle k \rangle_{\text{small}} = \sum_{k=1}^{\infty} k P(\text{degree } k | \text{small component})$ Average degree k in small components is then:

$$\langle k \rangle_{\text{small}} = \frac{1}{g_0(u)} \sum_{k=0}^{\infty} k p_k u^k = \frac{u g'_0(u)}{g_0(u)}$$

Average number of nodes reachable from an edge of a node i

$$\langle t \rangle = 1 + \langle k \rangle_{\text{neighbor}} \langle t \rangle \quad \rightarrow \quad \langle t \rangle = \frac{1}{1 - \langle k \rangle_{\text{neighbor}}}$$



$\langle k \rangle_{\text{neighbor}}$ is the average number of neighbors reachable from nodes like n_1, n_2, n_3 , excluding the link with node i , meaning it is also the average **excess degree** of nodes (i) in small components!!!

Excess degree distribution in small components is:

$$p(k_{\text{neighbor}}) = q(k|\text{small component}) = \frac{(k+1)p_{k+1}u^{k+1}}{\sum_{k+1} (k+1)p_{k+1}u^{k+1}} = \frac{(k+1)p_{k+1}u^{k+1}}{ug'_0(u)}$$

u^{k+1} is here to ensure small component

Average excess degree in small components

Average number of nodes reachable from an edge

$$\begin{aligned} \langle k \rangle_{\text{neighbor}} &= \frac{1}{ug'_0(u)} \sum_{k=0}^{\infty} k(k+1)p_{k+1}u^{k+1} = \frac{1}{ug'_0(u)} \sum_{k'=1}^{\infty} (k'-1)k'p_k u^{k'} \\ &= \frac{ug''_0(u)}{g'_0(u)}. \end{aligned}$$



$$\langle t \rangle = \frac{1}{1 - \frac{ug''_0(u)}{g'_0(u)}}$$

Average small component size

Average size of small components:

$$\langle s \rangle = 1 + \langle k \rangle_{\text{small}} \langle t \rangle$$

$$\langle k \rangle_{\text{small}} = \frac{u g'_0(u)}{g_0(u)}$$

$$\langle t \rangle = \frac{1}{1 - \frac{u g''_0(u)}{g'_0(u)}}$$

$$\langle s \rangle = 1 + \frac{u^2 g'_0(1)}{g_0(u) [1 - g'_1(u)]}$$

$$g_1(u) = \frac{g'_0(u)}{g_0(1)} \rightarrow g'_0(u) = g_1(u) g'_0(1) = u g'_0(1), \text{ and } g''_0(u) = g'_1(u) g'_0(1)$$

(Exercise)

When there is no giant component: $u = 1$ (also known as subcritical phase)

$$g_0(1) = \sum_k p_k = 1$$

$$\langle s \rangle = 1 + \frac{g'_0(1)}{1 - g'_1(1)}$$

Recall: $g_1(u) = \sum_k q_k u^k \rightarrow g'_1(u) = \sum_k k q_k u^{k-1} \rightarrow g'_1(1) = \sum_k k q_k = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}$ **is average excess degree**

At critical point: $\frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = 1 \rightarrow g'_1(1) = 1 \rightarrow \langle s \rangle \rightarrow \infty$ **Giant component is formed!!!**